

Project Proposal – CS3009 Software Engineering

Project Title: AI-Based Intelligent Load Balancing System for Dynamic Server Traffic Management

Course: CS3009 – Software Engineering

Team Members

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1. Introduction

Modern distributed systems and web applications handle thousands of simultaneous user requests. Efficient distribution of these requests across multiple servers is essential to maintain system performance, reliability, and scalability. Traditional load balancing techniques such as Round Robin and Least Connections rely on static rules and cannot adapt to dynamic traffic patterns or changing server conditions.

This project proposes the development of an AI-based intelligent load balancing system that predicts server load and dynamically routes incoming requests to the most suitable server. The system will simulate a server environment and compare traditional load balancing algorithms with an AI-driven routing approach to demonstrate improvements in response time and resource utilization.

2. Problem Statement

Traditional load balancing algorithms distribute requests without considering future traffic patterns or server performance trends. As a result, some servers may become overloaded while others remain underutilized. This can lead to increased response time and inefficient use of computing resources. A more intelligent system that can predict server load and dynamically adjust routing decisions is required.

3. Objectives

The main objectives of this project are:

- Design and simulate a server environment with multiple backend servers.

- Implement traditional load balancing algorithms such as Round Robin and Least Connections.
- Develop an AI-based prediction model that estimates server load based on traffic patterns.
- Integrate a Node.js load balancer with a Python-based AI prediction service.
- Compare the performance of traditional algorithms with the AI-based load balancing approach.
- Visualize system performance using a monitoring dashboard.

4. Proposed System

The proposed system will follow a modular architecture consisting of the following components:

1. Traffic Generator – Simulates incoming user requests.
2. Node.js Load Balancer – Distributes requests using multiple algorithms.
3. AI Prediction Service (Python) – Uses machine learning to predict server load and suggest optimal routing.
4. Server Pool – Multiple simulated backend servers processing requests.
5. Monitoring Dashboard – Displays system metrics such as request distribution, response time, and server utilization.

5. Software Engineering Concepts Applied

- Software Process Model: Agile development approach.
- Requirements Engineering: Functional and non-functional requirements documented in the SRS.
- System Modeling: UML diagrams including Use Case, Sequence, and Class diagrams.
- Architectural Design: Modular microservice-based architecture using Node.js and Python.
- Design Patterns: Strategy pattern for implementing different load balancing algorithms.
- Testing: Unit testing, integration testing, and performance testing.
- Project Management: Task scheduling using Gantt charts and risk management strategies.

6. Expected Outcomes

The final system will demonstrate how intelligent routing can improve server performance and resource utilization. The output will include a functional prototype, performance comparison between traditional and AI-based algorithms, a monitoring dashboard showing system metrics, and complete documentation including SRS and design diagrams.

7. Tools and Technologies

Backend Development: Node.js with Express.js

AI Prediction Service: Python with Scikit-learn

Visualization: Chart.js / Matplotlib

API Communication: REST APIs

Version Control: GitHub

8. Expected Contributions

This project will demonstrate how AI techniques can be integrated with modern backend architecture to improve performance and scalability. The system will provide insights into intelligent traffic routing and resource optimization in distributed systems.